

5.(a)	Draw the working of CRO with proper diagram. Write the measurements which can be done using CRO.	6	5
(b)	Draw the diagram of Digital Storage Oscilloscope and describe its working in brief.	4	5

Q4(a)	The expected value of the current through a resistor is 20 mA. However the measurement yields a current value of 18 mA. Calculate (i) absolute error (ii) % error and (iii) relative accuracy	3	CO4	K5																						
Q4(b)	The below table gives the set of ten measurement that were recorded in the laboratory. Calculate the precision of 5th measurement. <table><tr><th>Measurement Number</th><th>Measurement Value</th></tr><tr><td>1.</td><td>98</td></tr><tr><td>2.</td><td>101</td></tr><tr><td>3.</td><td>102</td></tr><tr><td>4.</td><td>97</td></tr><tr><td>5.</td><td>101</td></tr><tr><td>6.</td><td>100</td></tr><tr><td>7.</td><td>103</td></tr><tr><td>8.</td><td>98</td></tr><tr><td>9.</td><td>106</td></tr><tr><td>10.</td><td>99</td></tr></table>	Measurement Number	Measurement Value	1.	98	2.	101	3.	102	4.	97	5.	101	6.	100	7.	103	8.	98	9.	106	10.	99	2	CO4	K5
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Q4(c)	Explain the construction and working of LVDT.	5	CO4	K2																						

4(a)	Discuss the principle of operation of a Linear Variable Differential Transformer (LVDT). Also, state its advantages and disadvantages.	5	CO4	K2, K4
4(b)	What is transducer? Explain the difference between active and passive transducer with examples.	5	CO4	K1,

4.(a)	Show the working of LVDT with proper diagrams.	5	4
(b)	Expected value of voltage across a resistor is 90 V. However measured value is 88.5 V. Calculate (i) Percentage Error (ii) Relative Accuracy	2	4
(c)	Explain the functional block diagram of an instrument.	3	4